Using Low Dropout Regulators For Consumer Electronics

To keep up with recent demands for linear regulators in consumer electronic equipment, overall performance of low dropout regulators (LDO) regulators, such as power conversion performance, additional functions to adapt themselves into consumer electronics easily and reliability are being closely examined and proven

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Various kinds of integrated circuits (ICs) are now indispensable components in almost all fields of the present human life. Consequently, power management for these ICs is getting important for electronic devices, including consumer electronics, those for industrial applications, battery-powered electronics and many others that require a reliable and efficient power management solution.

Key requirements of power management are also quite obvious for consumer electronics, such as stable power delivery, less power consumption under various load conditions, low noise, less space, high reliability and so on. As a result, various power management ICs—such as linear regulators, DC-DC converters, power controllers and PMICs—have been developed and applied to meet the specific requirements of various application fields.

Low dropout regulators (LDOs) are a simple, inexpensive power component to regulate output voltage powered from a higher voltage input. Advantages of these are easy-to-design power circuit and decent power quality. For most applications, performance index of LDOs, is simple and clear to understand and apply.

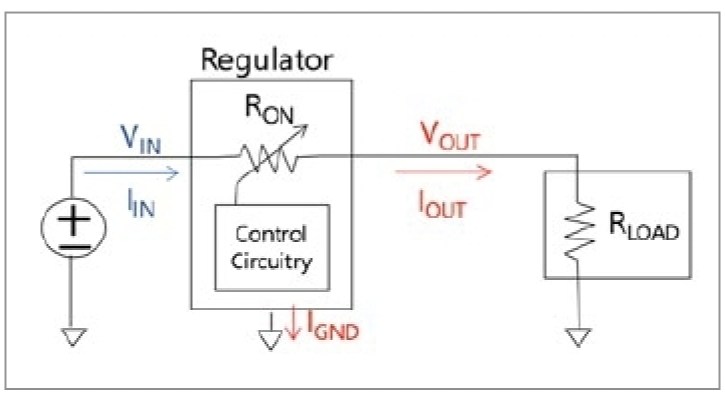

Fig. 1: Operating concept of a linear regulator

Dropout voltage

Linear regulators can be classified by standard linear regulators, low dropout linear regulators (LDOs) and ultra-low dropout linear regulators (ULDOs). The key difference among these is the dropout voltage characteristic that is required to maintain a regulated output voltage.

Dropout voltage is defined as the minimum input to output differential voltage at which output maintains regulation. As a viewpoint of power delivery, a linear regulator can be regarded as a variable resistor, as shown in Fig. 1.

When I_{OUT} is changed (R_{LOAD} is changed), internal control circuitry of the linear regulator adjusts on-resistance of pass element. So, adjustable on-resistance range defines the operating maximum current range (operating output impedance range). For standard linear regulators, pass element is either a Darlington NPN or PNP output stage, and these have dropout voltages as high as 2V. This kind of linear regulators cannot be applied to applications that need lower dropout voltages, such as generating 3.3V from 3.6V battery power. To get lower dropout voltage characteristics, most LDOs/ULDOs use NMOSFET or PMOSFET of appropriate size as a pass element.

The important reason why the power circuit should pay attention to dropout voltage is power conversion efficiency. Lower dropout voltage performance of an LDO makes lower input voltage supply possible for the required output power delivery. This results in higher power conversion efficiency and lower heat occurrence problems due to power dissipation.